Sleep Quality Estimation with Adversarial Domain Adaptation: From Laboratory to Real Scenario

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Abstract*—Previous studies on EEG-based last-night sleep quality estimation mainly focus on evaluation with data from laboratory experiments. However, due to the reality gap constituted with device performance, subject groups, experiment settings and controlled conditions, the models trained solely on laboratory data cannot generalize well to real scenarios. In this work, we investigate the sleep quality estimation for high-speed train drivers as an instance of real-scenario application. Domain adaptation models are adopted to deal with the individual differences across subjects when modeling and testing with the real-scenario data. As it is usually difficult and costly to acquire data and annotate them in real scenarios, the high-quality data in laboratory conditions are used for model trainings. Knowledge from simulation is transferred to reality with domain adaptation methods. A novel approach called Domain Adversarial Neural Network (DANN) is adopted. DANN learns domain independent features through deep networks with an adversarial architecture. The experimental results indicate that DANN outperforms other state-of-the-art methods and achieves 19.55% and 23.50% improvements in terms of accuracy on the cross-subject and cross-scenario tasks, respectively, in comparison with the baseline SVM model.

I. Introduction

Sleep has always been an important topic, no matter in research field or in daily life. Insufficient sleep may lead to serious impairment in daytime performance, increase the risk of driving and occupational accidents and result in diminished life quality [1]. Akerstedt *et al*. studied the relationship between sleepiness and accidents in transport operation. They concluded that sleepiness causes 15% to 20% of all accidents, surpassing alcohol- or drug-related incidents in all modes of transportation, becoming the largest identifiable and preventable cause of accidents [2]. Besides, driver sleepiness has also become a major threat to the railway safety. Bad sleep quality affects drivers' attention, judgment and execution, which may trigger railway fatalities and gravely damage the security of people's lives and properties [3]. Therefore, a reliable measurement of sleep quality, especially last-night sleep quality, becomes necessary.

Currently, there are subjective approaches judging sleep quality by self-evaluation via questionnaires, interviews and sleep dairy. These self-evaluation approaches usually require the participants to consider their sleep habits of the recent times. Pittsburgh Sleep Quality Index (PSQI) [4] and Epworth Sleep Scale (ESS) [5] are representative ones widely adopted by sleep researchers. PSQI limits the time interval to the last month and ESS refers to the participant's usual way of life in recent times. Taking considerably long time into account makes the measurement more precise and robust in distinguishing patients from healthy people. However, these subjective approaches would fail when evaluating the last-night sleep quality due to the constrains of taking a relatively long period into account.

Various objective approaches have been proposed to precisely measure the sleep quality over a single night. Polysomnography (PSG) [6] is a typical objective method of this kind. The PSG monitors body functions including brain, eye movements, muscle activity, skeletal muscle activation and heart rhythm during sleep. The recorded signals are then analyzed by a doctor who is capable of judging the sleep quality. This approach requires a PSG device attached to the subject during the whole sleeping process, which makes it not an applicable choice in real scenarios. For instance, if PSG is applied to high-speed train driver sleep quality checking, the company would have to purchase a PSG device for every driver, which would be rather expensive, and the sleep quality checking procedure would also be time-consuming and inefficient. Therefore, a cheaper and easier method is in demand.

With the rapid development of wearable electroencephalography (EEG) signal acquiring devices, EEG-based last-night sleep quality measurement is considered as a feasible choice. This approach does not require EEG during the whole sleep process, but only acquires EEG in a short time after the subject wakes up. Ideally, these EEG data would be containing information of the subject's last-night sleep quality. Most EEG-based sleep studies are done in the laboratory environment with precisely controlled sleep duration or sleep deprivation. With different controlled conditions, such as restricting sleep time to be 4, 6 or 8 hours [7], the subjects would be in totally different mental states, which makes it easier to distinguish

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睡眠质量估计与对抗域适应：从实验室到真实场景

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摘要—基于脑电图（EEG）的昨晚睡眠质量评估的先前研究主要集中于使用实验室实验数据进行的评估。然而，由于设备性能、受试者群体、实验设置和受控条件所构成的现实差距，仅使用实验室数据训练的模型无法很好地泛化到真实场景。在这项工作中，我们以高速列车驾驶员为例，研究了真实场景应用的睡眠质量评估。采用领域自适应模型来处理在用真实场景数据建模和测试时受试者之间的个体差异。由于在真实场景中获取数据和标注通常既困难又昂贵，因此使用实验室条件下的高质量数据进行模型训练。通过领域自适应方法将模拟知识转移到现实。采用了一种称为领域对抗神经网络（DANN）的新方法。DANN 通过具有对抗架构的深度网络学习领域无关特征。 实验结果表明，DANN 比其他最先进的方法表现更优，在跨被试和跨场景任务上，相较于基线 SVM 模型，准确率分别提高了 19.55%和 23.50%。

I. 睡眠质量始终是一个重要话题，无论在研究领域还是日常生活中。睡眠不足可能导致白天表现严重受损，增加驾驶和职业事故的风险，并降低生活质量[1]。Akerstedt 等人研究了运输操作中嗜睡与事故的关系。他们得出结论，嗜睡导致所有事故的 15%至 20%，超过所有交通方式中酒精或药物相关的事故，成为最大的可识别和可预防的事故原因[2]。此外，驾驶员的嗜睡也成为铁路安全的主要威胁。睡眠质量差会影响驾驶员的注意力、判断力和执行力，可能引发铁路事故，严重损害人民生命财产安全[3]。因此，对睡眠质量进行可靠测量，尤其是昨晚的睡眠质量，变得十分必要。

目前，存在主观方法通过问卷调查、访谈等方式自我评估睡眠质量，睡眠日记。这些自我评估方法通常要求参与者考虑他们最近时期的睡眠习惯。匹兹堡睡眠质量指数（PSQI）[4]和爱泼沃斯睡眠量表（ESS）[5]是睡眠研究人员广泛采用的代表性方法。PSQI 将时间间隔限制在最近一个月，ESS 则参考参与者在最近时期的通常生活方式。考虑到较长的时间间隔使得测量更加精确，并且在区分患者和健康人方面更加稳健。然而，由于考虑相对较长的时间段，这些主观方法在评估昨晚睡眠质量时会失效。

已经提出了多种客观方法来精确测量单夜的睡眠质量。多导睡眠图（PSG）[6] 是这类方法中的一种典型客观方法。PSG 监测睡眠期间的身体功能，包括大脑、眼动、肌肉活动、骨骼肌激活和心率。记录的信号随后由能够判断睡眠质量的医生进行分析。这种方法需要在整个睡眠过程中将 PSG 设备连接到受试者身上，这使得它不适用于实际场景。例如，如果将 PSG 应用于高速列车司机睡眠质量检查，公司将不得不为每位司机购买 PSG 设备，这将相当昂贵，而且睡眠质量检查过程也会耗时且效率低下。因此，需要一种更便宜、更简单的方法。

随着可穿戴脑电图（EEG）信号采集设备的快速发展，基于 EEG 的昨晚睡眠质量测量被认为是一种可行的选择。这种方法不需要在整个睡眠过程中都进行 EEG 监测，而只需在受试者醒来后短时间内采集 EEG。理想情况下，这些 EEG 数据应包含受试者昨晚睡眠质量的信息。大多数基于 EEG 的睡眠研究都是在实验室环境中进行的，睡眠时长或睡眠剥夺都经过精确控制。在不同的控制条件下，例如将睡眠时间限制为 4、6 或 8 小时[7]，受试者的心理状态将完全不同，这使得区分

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between different sleep quality with EEG signals. These studies achieved promising results on restricted laboratory environments. However, to the best of our knowledge, no studies under real scenarios has been performed yet.

In this paper, we aim to apply the EEG-based last-night sleep quality estimation approach to real scenarios. To acquire real-scenario data, we perform experiments on high-speed train drivers with different last-night sleep qualities at their workplace. We first perform a cross-subject task on real-scenario data. Domain adaptation (DA) methods are adopted to reduce the individual difference across subjects. As the data collection and annotation in real scenarios are usually costly and difficult, using only the real-scenario data to train models is not practicable. We introduce the laboratory data used our in previous work and perform a cross-scenario task in which we train models on laboratory data and test them on real-scenario data. DA methods are also adopted to narrow the reality gap between these two scenarios. Among the adopted DA methods, the Domain Adversarial Neural Network (DANN) method performs the best in terms of classification accuracy with considerable improvement over the other methods.

II. Related Work

Sleep has been a hot research topic for decades. In the early days, sleep is defined as a human maitaining a specific body gesture and behavioral quiescence with elevated arousal threshold and state reversibility after stimulation [8].

Since the development of technology to amplify and record spontaneous EEG signals, it has been demonstrated that there is a strong correlation between EEG and sleep. Hori *et al*. proposed a nine-stage EEG-based system for identifying successive EEG changes throughout the sleep onset period which represents the greatest advance in understanding the sleep onset process yet achieved [8]. Wolpert *et al*. proposed a sleep stage character system depending on brain waves [9], in which a distinct and typical EEG and physiologic patterns for alert, REM sleep and each stage of NREM has been found. Therefore, researchers in sleep field have always been paying close attention to EEG signals.

In the last decade, the differences and changes of EEG under different sleep qualities have been investigated. Li *et al*. tried to study how lack of sleep influences the event-related potentials under stimulation [10]. Audio stimulation was given to subjects in both cases of sufficient sleeping and lack of sleep and event-related potentials of the subjects were analyzed with the parallel factor analysis method. They found that when subjects are lack of sleep, event-related potentials activities of the subjects tend to appear near to the forehead and Gamma frequency band delay and attenuation would occur. Na *et al*. investigated effects of sleep deprivation on the connection between cerebral hemisphere. They conducted a mutual information analysis on the EEG acquired from normal sleep and sleep deprived subjects and concluded that the connection between cerebral hemisphere was weaker in case of sleep deprivation [11]. Tassi *et al*. focused on the correlation between partial sleep deprivation and the EEG activity afterward. The spectral analysis applied on the waking EEG during cognitive performance testing shows that alpha activities were increased in both deprived and normal subjects but theta power increased only in the sleep deprived group [12]. The studies mentioned above mainly focus on the comparison between controlled experiment of two cases: the deprived group and the normal group. They usually analyze EEG data in a statistical manner and achieve qualitative conclusions.

In recent years, a minority of researchers take up research into quantitative last-night sleep quality estimation with machine learning methods. Wang *et al*. attempt to measure last-night sleep quality from resting EEG signals [3]. They designed an experiment collecting resting EEG signals from subjects under three discrete sleep conditions: 8 hours sleep, 6 hours sleep, and 4 hours sleep, according to which the EEG data are labeled as three categories. To correctly classify the EEG data, they introduced discriminative graph regularized extreme learning machine together with minimal-redundancy-maximal-relevance feature selection algorithm and achieved a mean accuracy of 83.5% in a leave-one-subject-out validation manner [3]. Zhang *et al*. followed that work and modified the EEG acquisition procedure, which we discuss in the experiment section [7]. They extracted subject independent features with three domain adaptation methods and feed them into SVM to make comparisons. With considerable model performance improvement, they concluded that domain adaptation approaches do have the capability to reduce the differences of EEG data across subjects and sessions [7].

In this work, we focus on last-night sleep quality estimation in real scenarios. In our previous work [3] [7], EEG signals are acquired with wet-electrode devices, which is not applicable to real scenarios due to the laborious preparation process and poor portability of wet-electrode devices. Therefore, we use a dry-electrode device to perform EEG acquiring experiments on high-speed train drivers. We also adapt our experiment settings to the real-scenario conditions. Differences of devices, subject groups, experiment settings and controlled conditions lead to a giant reality gap between laboratory and real-scenario data, making the last-night sleep quality estimation in real scenario a much more challenging task than in laboratory conditions. To address the domain difference problem, we adopt not only traditional DA methods but also a novel DANN method with the capability of learning domain independent features through deep neural networks with an adversarial architecture. DANN outstandingly outperforms other models and turns out to be stable and robust according to our experimental results.

III. Methods

A. Feature Extraction

EEG signals can be divided into five different frequency bands: Delta (1∼4 Hz), Theta (4∼8 Hz), Alpha (8∼13 Hz), Beta (13∼30 Hz) and Gamma (30∼50 Hz) bands [13]. The feature extraction procedure converts the time domain signals to the frequency domain and then extracts useful information for the five frequency bands. Differential entropy (DE) features

在 EEG 信号方面，不同睡眠质量之间的研究已经取得了在受限实验室环境中的良好成果。然而，据我们所知，目前还没有在真实场景下进行过相关研究。

在本文中，我们旨在将基于脑电图（EEG）的昨晚睡眠质量评估方法应用于实际场景。为了获取实际场景数据，我们在高速列车驾驶员的工作场所对不同昨晚睡眠质量的驾驶员进行实验。我们首先在真实场景数据上执行跨被试任务。采用领域自适应（DA）方法来减少被试间的个体差异。由于在实际场景中收集和标注数据通常成本高昂且困难，仅使用真实场景数据来训练模型并不切实际。我们引入了先前工作中使用的实验室数据，并执行跨场景任务，其中我们在实验室数据上训练模型，并在真实场景数据上测试模型。同样采用 DA 方法来缩小这两个场景之间的现实差距。在所采用的 DA 方法中，领域对抗神经网络（DANN）在分类精度方面表现最佳，相较于其他方法有显著提升。

II. RW

睡眠几十年来一直是热门的研究课题。早期，睡眠被定义为人类保持特定的身体姿势和行为静止，在受到刺激后具有升高的唤醒阈值和状态可逆性[8]。

自从开发出放大和记录自发脑电图信号的技术以来，已经证明脑电图与睡眠之间存在很强的相关性。Hori 等人提出了一种基于脑电图的九阶段系统，用于识别睡眠开始期间连续的脑电变化，这是迄今为止对睡眠开始过程理解的最大进步[8]。Wolpert 等人提出了一种基于脑电波的睡眠阶段特征系统[9]，其中发现了对清醒、快速眼动睡眠和每个非快速眼动睡眠阶段的独特且典型的脑电图和生理模式。因此，睡眠领域的研究者一直密切关注脑电图信号。

在过去的十年中，研究人员已经对在不同睡眠质量下脑电图（EEG）的差异和变化进行了研究。李等人试图研究睡眠不足如何影响刺激下的事件相关电位[10]。在充足睡眠和睡眠不足两种情况下，研究人员对受试者进行音频刺激，并使用平行因子分析方法分析受试者的事件相关电位。他们发现，当受试者睡眠不足时，事件相关电位活动倾向于出现在额头附近，并且伽马频段会出现延迟和衰减。Na 等人研究了睡眠剥夺对大脑半球连接的影响。他们对正常睡眠和睡眠剥夺受试者获得的脑电图进行了互信息分析，并得出结论认为，在睡眠剥夺的情况下，大脑半球的连接会减弱[11]。Tassi 等人专注于部分睡眠剥夺与之后脑电图活动之间的相关性。在认知性能测试期间对清醒时的脑电图进行的频谱分析表明，在睡眠剥夺组和正常组中，alpha 活动均增加，但只有睡眠剥夺组的 theta 功率增加[12]。上述研究主要关注两种情况（睡眠剥夺组和正常组）的对照实验比较。它们通常以统计方式分析脑电图数据，并得出定性的结论。

近年来，少数研究人员开始采用机器学习方法研究定量化的昨晚睡眠质量评估。王等人尝试从静息脑电信号中测量昨晚睡眠质量[3]。他们设计了一个实验，收集受试者在三种离散睡眠条件下的静息脑电信号：8 小时睡眠、6 小时睡眠和 4 小时睡眠，根据这些条件将脑电数据标记为三类。为了正确分类脑电数据，他们引入了判别图正则化极限学习机以及最小冗余最大相关特征选择算法，并在留一受试者验证方式下达到了 83.5%的平均准确率[3]。张等人随后进行了这项工作，并修改了脑电采集程序，我们将在实验部分进行讨论[7]。他们使用三种域适应方法提取受试者独立特征，并将其输入支持向量机进行比较。通过显著提升模型性能，他们得出结论：域适应方法确实能够减少受试者之间和不同会话中脑电数据的差异[7]。

在这项工作中，我们专注于真实场景中昨晚睡眠质量的评估。在我们的先前工作中[3][7]，通过湿电极设备采集脑电图信号，但由于湿电极设备的准备过程繁琐且便携性差，不适用于真实场景。因此，我们使用干电极设备对高速列车司机进行脑电图采集实验，并将实验设置调整到真实场景条件。设备差异、受试者组别、实验设置和受控条件的不同，导致实验室和真实场景数据之间存在巨大的现实差距，使得真实场景中的昨晚睡眠质量评估比实验室条件下的任务更具挑战性。为了解决领域差异问题，我们不仅采用了传统的领域自适应方法，还采用了一种具有通过具有对抗架构的深度神经网络学习领域无关特征能力的新型 DANN 方法。根据我们的实验结果，DANN 显著优于其他模型，并表现出稳定性和鲁棒性。

III. M

A. 特征提取

脑电图信号可以分为五个不同的频段：Delta（1~4 Hz）、Theta（4~8 Hz）、Alpha（8~13 Hz）、Beta（13~30 Hz）和 Gamma（30~50 Hz）频段[13]。特征提取过程将时域信号转换为频域，然后提取五个频段的有用信息。微分熵（DE）特征

are commonly used for its reflecting the energy change of EEG signals [14]. We extracted DE features from the 18-channel EEG signals in 5 frequency bands, which add up to a feature vector of length 90. Since the extracted features contain fluctuations caused by noise, we use the linear dynamical system to remove those fluctuations [15].

B. Domain Adaptation models

Domain adaptation helps to transfer knowledge from a source domain to a different but related target domain, even though these domains may have different distributions. Most existing DA methods aim at finding a new feature representation reducing the difference between the distributions of the source and target domains, meanwhile preserving the data properties of the source and target domains. In this work, we would apply DA methods to transfer knowledge from our laboratory experiments to the real-scenario (high-speed train drivers) experiments. Below, we introduce the eight methods adopted in this work.

1) Transfer Component Analysis (TCA) [16]*:* TCA aims to learn a transformation that reduces the distance between the marginal distributions and preserves the important properties of both domains. TCA empirically measures the distance between the source and target distributions by the distance between the empirical means of the two domains in Reproducing Kernel Hilbert Space (RKHS) . Besides, to preserve the main properties of both domains, the TCA chooses to maximally preserve their variance.

2) Information-Theoretical Learning (ITL) [17]*:* ITL assumes that in a latent feature space induced by a linear transformation $\mathbf{L} \in \mathbb{R}^{d \times D}$, data in source and target domains are clustered according to their labels and the clusters from two domains that possess the same label are geometrically close . Then a k-nearest neighbors model is used to estimate the label of each sample in the target domain. The negated mutual information between the target data and the estimated label is used to approximate the classification error, which is to be minimized. Another k-nearest neighbors model is used to estimate the domain label of each sample in both domains. To express the desideratum that clusters possessing the same label are geometrically close, the negated mutual information between samples and domain labels is maximized.

3) Geodesic Flow Kernel (GFK) [18]*:* GFK parameterizes the process of the source domain smoothly changing to the target domain with a geodesic flow $\mathbf{\Phi}(t)$, where $t \in [0, 1]$. $\mathbf{\Phi}(t)$ smoothly changes from the basis of subspace for the source domain to that of the target domain when t gradually changes from 0 to 1. GFK then expands the original feature $\mathbf{x}$ by projecting it onto all these subspaces, which gives us an infinite-dimensional feature vector $\mathbf{z}^{\infty}$. Then the inner product in this infinite dimensional space would be

$$\langle \mathbf{z}_i^{\infty}, \mathbf{z}_j^{\infty} \rangle = \int_0^1 \left(\mathbf{\Phi}(t)^T \mathbf{x}_i\right)^T \left(\mathbf{\Phi}(t)^T \mathbf{x}_j\right) = \mathbf{x}_i^T \mathbf{G} \mathbf{x}_j. \quad (1)$$

This is a "kernel trick", and the matrix $\mathbf{G}$ has a closed form solution.

4) Joint Distribution Adaptation (JDA) [19]*:* JDA hopes to find projection matrix $\mathbf{A}$ that adapts the joint distributions and the conditional distribution of features $\mathbf{x}$ and labels $\mathbf{y}$ simultaneously. Since there is no labeled data in the target domain, it uses a classifier f to generate pseudo target label as an approximation of actual label. In order to achieve more accurate approximation, an iterative pseudo label refinement strategy is introduced to refine the transformation and the classifier.

5) Subspace Alignment (SA) [20]*:* SA represents the source and target domains as subspaces described by eigenvectors. It then seeks a mapping function that aligns the source and target subspaces as a domain adaptation solution. Using PCA, it selects d eigenvectors for each domain as their corresponding source ($\mathbf{X}_S$) and target ($\mathbf{X}_T$) subspaces. Then, a linear transformation matrix $\mathbf{M}$ is used to align $\mathbf{X}_S$ with $\mathbf{X}_T$. $\mathbf{M}$ is learned by minimizing the Bregman matrix divergence.

6) Transfer Joint Matching (TJM) [21]*:* TJM proposes to adapt domains with a transformation T that (a) minimize the *Maximum Mean Discrepancy* distance between the source and target domains, and (b) reweight the source instances through a structured sparsity. TJM works in RKHS to match both first- and high-order statistics. As there is a row-sparsity regularizer in the optimization objective, TJM solves the problem by an iterative approach.

7) Maximum Independence Domain Adaptation (MIDA) [22]*:* MIDA shares the same intuition with TCA. It aims at minimizing the domain difference while preserving the data properties with transformation Φ. With kernel trick, the kernel $\mathbf{K}$ matrix can be easily induced. A projection matrix $\tilde{\mathbf{W}}$ is used to transform the kernel map. MIDA assumes that $\tilde{\mathbf{W}} = \Phi(\mathbf{X})\mathbf{W}$. MIDA measures the dependence between projected samples and domain features by Hilbert-Schmidt independence criterion (HISC). Besides, MIDA preserves the properties of data by keeping the variance in the transformed feature space.

8) Domain Adversarial Neural Network (DANN) [23]*:* DANN is a domain adaptation approach with deep architectures that can be trained with labeled source domain data and unlabeled target domain data. Its adaptation behavior is achieved by augmenting a normal feed-forward model with a few standard layers that work as a domain discriminator and a gradient reversal layer that makes the model be trained procedure in an adversarial manner. Figure 1 depicts the architecture we adopt in this work.

DANN estimates the dissimilarity of domains in feature representation f by looking at the loss of domain classifier G_d. Therefore, at training time, DANN seeks the parameters θ_f that maximize the loss of G_d and meanwhile, seeks for the parameters θ_d that minimize the loss of G_d. By introducing a gradient reversal layer (GRL) $R_\lambda(\cdot)$ between the feature f and the domain classifier G_d, one can define GRL as a pseudo-function by its forward- and back-propagation behaviors:

$$R_\lambda(f) = f \quad (2)$$

$$\frac{\partial R_\lambda(f)}{\partial f} = -\lambda I, \quad (3)$$

通常用于反映脑电图信号的能量变化[14]。我们从 5 个频段的 18 通道脑电图信号中提取 DE 特征，构成一个长度为 90 的特征向量。由于提取的特征包含由噪声引起的波动，我们使用线性动力系统来去除这些波动[15]。

B. 领域自适应模型

领域自适应有助于将知识从源域迁移到不同的但相关的目标域，即使这些域可能有不同的分布。大多数现有的 DA 方法旨在找到一个新的特征表示，以减少源域和目标域分布之间的差异，同时保留源域和目标域的数据特性。在这项工作中，我们将应用 DA 方法将知识从我们的实验室实验迁移到真实场景（高速列车驾驶员）实验。下面，我们介绍了这项工作中采用的八种方法。

1) 转移成分分析 (TCA) [16]: TCA 旨在学习一个变换，以减少边缘分布之间的距离并保留两个域的重要特性。TCA 通过在再生核希尔伯特空间 (RKHS) 中测量两个域的经验均值之间的距离来经验性地测量源域和目标域之间的距离。此外，为了保留两个域的主要特性，TCA 选择最大限度地保留它们的方差。

2) 信息论学习 (ITL) [17]: ITL 作为假设在线性变换 $L \in R$ 诱导的潜在特征空间中，源域和目标域中的数据根据其标签进行聚类，并且具有相同标签的两个域的聚类在几何上接近。然后使用 k 近邻模型来估计目标域中每个样本的标签。目标数据和估计标签之间的负互信息用于近似分类误差，该误差需要被最小化。另一个 k 近邻模型用于估计两个域中每个样本的域标签。为了表达具有相同标签的聚类在几何上接近的要求，样本和域标签之间的负互信息被最大化。

3) 地形流核 (GFK) [18]: GFK 参数化源域通过地测流 $\Phi(t)$ 平滑地变化到目标域的过程，其中 $t \in [0, 1]$。当 t 从 0 逐渐变化到 1 时，$\Phi(t)$ 从源域子空间的基平滑地变化到目标域的基。GFK 然后将原始特征 x 投影到所有这些子空间上，从而得到一个无限维的特征向量 z。然后在这个无限维空间中的内积将是

$$\langle z, z\rangle = \int_0 (\Phi(t)x)(\Phi(t)x) = xGx. \quad (1)$$

这是一种“核技巧”，矩阵 G 有封闭形式的解。

4) 联合分布适应 (JDA) [19]: JDA 希望寻找投影矩阵 A，以同时适应特征 x 和标签 y 的联合分布和条件分布。由于目标域中没有标记数据，它使用分类器 f 生成伪目标标签作为实际标签的近似。为了实现更精确的近似，引入了一种迭代伪标签细化策略来优化转换和分类器。

5) 子空间对齐 (SA) [20]: SA 将源域和目标域表示为特征向量描述的子空间。然后，它寻找一个映射函数，将源域和目标域子空间对齐，作为域适应解决方案。使用 PCA，它为每个域选择 d 个特征向量，作为其对应的源 (X) 和目标 (X) 子空间。然后，使用线性变换矩阵 M 对齐 X 和 X。M 通过最小化 Bregman 矩阵散度来学习。

6) 转移联合匹配 (TJM) [21]: TJM 提出使用变换 T 来适应域，该变换 (a) 最小化源域和目标域之间的最大均值差异距离，以及 (b) 通过结构化稀疏性重新加权源实例。TJM 在再生核希尔伯特空间中工作，以匹配第一阶和高级统计量。由于优化目标中存在行稀疏性正则化器，TJM 通过迭代方法解决问题。

7) 最大独立性域适应 (MIDA) [22]: MIDA 与 TCA 分享相同的直觉。它旨在通过变换 Φ 最小化域差异，同时保留数据属性。通过核技巧，核矩阵 K 可以轻松诱导。使用投影矩阵

$\tilde{W}$ 来变换核映射。MIDA 假设 $\tilde{W} = \Phi(X)W$。MIDA 通过希尔伯特-施密特独立性准则 (HISC) 衡量投影样本与域特征之间的依赖关系。此外，MIDA 通过在变换的特征空间中保持方差来保留数据的属性。

8) 域对抗神经网络 (DANN) [23]: DANN 是一种具有深度架构的领域适应方法，可以使用标记的源域数据和未标记的目标域数据进行训练。它的适应行为是通过在正常的前馈模型中增加几个标准层来实现的，这些层作为领域判别器，以及一个梯度反转层，使模型以对抗方式进行训练。图 1 描绘了我们在这项工作中采用的架构。

DANN 通过观察领域分类器 G 的损失来估计特征表示 f 中域的差异性。因此，在训练时，DANN 寻求最大化 G 的损失的参数θ，同时寻求最小化 G 的损失的参数θ。通过在特征 f 和领域分类器 G 之间引入一个梯度反转层（GRL）R(・)，可以将其定义为伪函数，根据其前向和反向传播行为：

$$R(f) = f \quad (2)$$

$$\frac{\partial R(f)}{\partial f} = -\lambda I, \quad (3)$$

where λ is a tradeoff parameter. The objective function can then be optimized with statistic gradient descent (SGD) and also other optimization methods based on SGD.

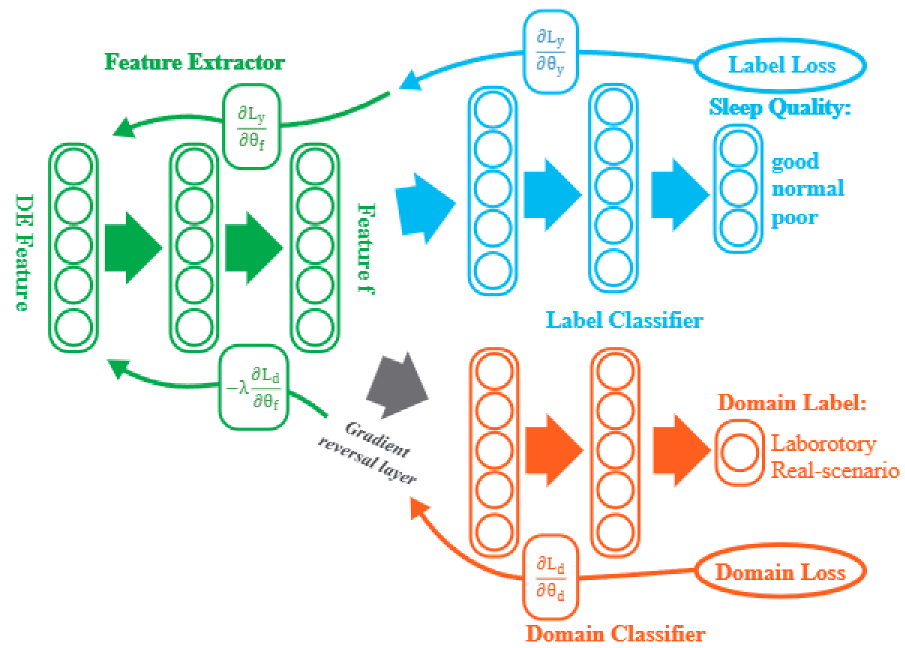


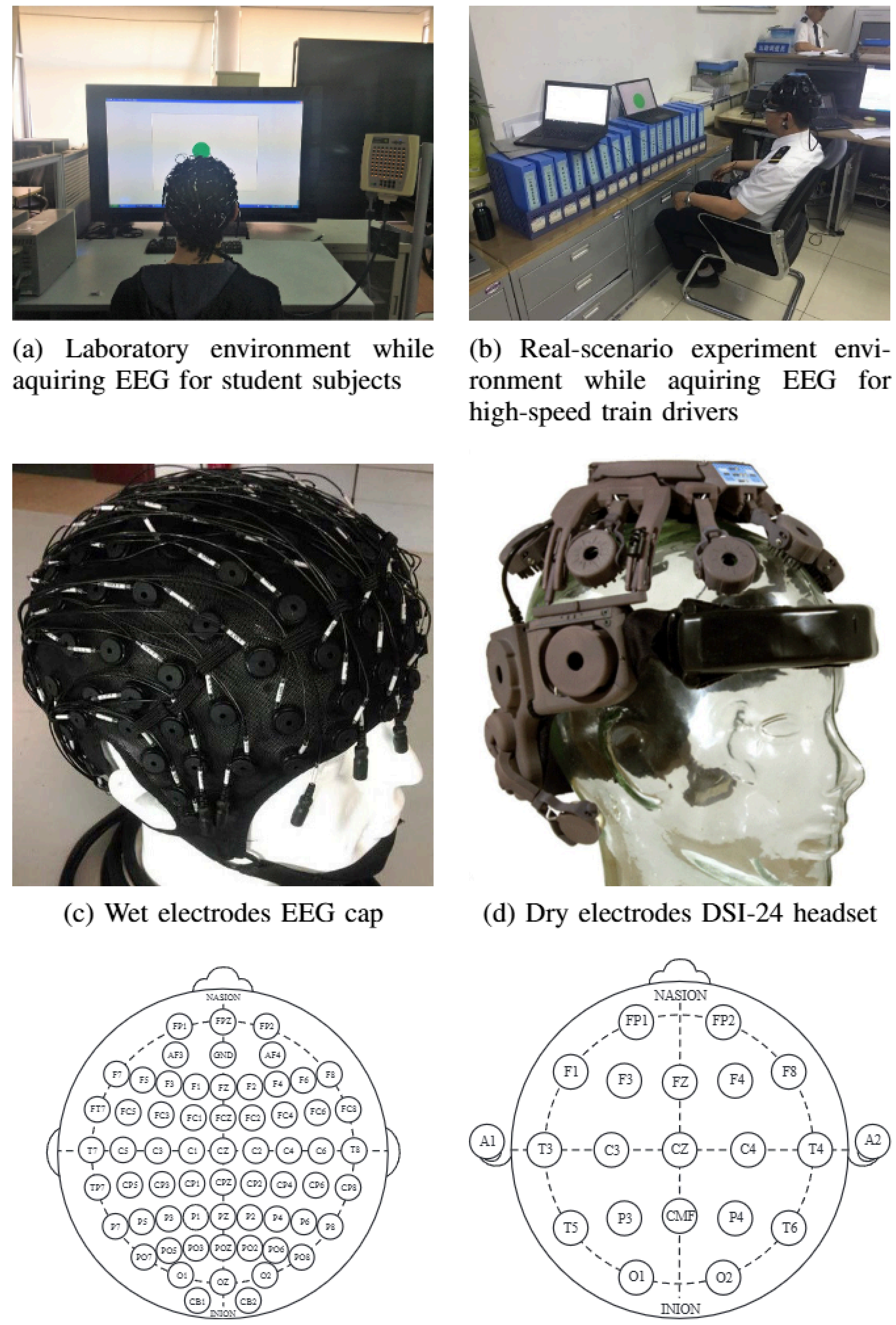
Fig. 1. The DANN model with three components: a feature extractor G_f, a label predictor G_y and a domain classifier G_d. The feature extractor maps the input x to a feature vector $f \in \mathbb{R}^m$ with all of its layer parameters denoted as θ_f, *i.e.* $f = G_f(x; \theta_f)$. The label predictor maps f to a predicted label $\hat{y}$ with all of its layer parameters denoted as θ_y. The domain classifier maps f to a predicted domain mark $\hat{d}$ with parameters denoted as θ_d.

IV. Experiments

Since we aim at transferring knowledge from the laboratory to the real scenario, it is necessary to have data from both scenarios. The laboratory data used in this work are acquired in our previous studies of sleep quality estimation [7], and in this work, we collected the real-scenario data from high-speed train drivers. Here, for comparison between these two types of experiments and easy understanding of how different the real-scenario data is from the laboratory ones, we will describe both of them in details. The data are preprocessed after acquisition and DE features are extracted from the preprocessed data. We then perform the cross-subject and cross-scenario tasks on the extracted features. A baseline SVM model and eight DA methods are then adopted in both tasks.

A. Laboratory Sleep Experiment

The Laboratory experiment aims at collecting resting EEG after sleep of different quality. Each experiment consists of two parts: sleep and EEG acquisition. 10 subjects (six males and four females, age range: 21-26, mean: 23.57, std: 1.62) were recruited for the experiments. At the night before EEG acquisition, the subject was required to sleep for a specified amount of time: 4, 6 or 8 hours corresponding to a good, normal or poor sleep quality [3]. To ensure that the subject slept for the time as required, a smart band was attached to them monitoring their wrist motions, which can rarely be detected when human fall asleep [7]. After the subject woke up, the EEG acquisition procedure started in less than an hour and lasted for 30 minutes. The EEG signals were recorded with a 62-channel wet electrode cap using the ESI NeuroScan system. Electrodes on the cap are placed according to the higher-resolution international 10-20 system. The left column of Figure 2 shows the experiment scene and the wet electrode device. During the data acquisition procedure, the subjects were required to stare at a green dot on the screen and count silently to keep a peaceful and concentrated state [7].



(a) Laboratory environment while aquiring EEG for student subjects

(b) Real-scenario experiment environment while aquiring EEG for high-speed train drivers

(c) Wet electrodes EEG cap

(d) Dry electrodes DSI-24 headset

(e) Higher-resolution international 10-20 system

(f) Normal international 10-20 system

Fig. 2. Experiment details of laborotary (left column) and real-scenario (right column) experiments.

B. Real-Scenario Sleep Experiment

TABLE I
SLEEP DURATION AND CORRESPONDING SLEEP QUALITY

Sleep time	Sleep quality	Subjects
(2, 5] h	Poor	10
(5, 7] h	Normal	24
(7, 10] h	Good	36

The real-scenario experiments are performed to record the EEG data for subjects from a specific group. The subjects should be going through their daily routine and not be restricted by any condition set by the experiment. In this work,

其中 λ 是权衡参数。目标函数可以使用统计梯度下降（SGD）以及其他基于 SGD 的优化方法进行优化。

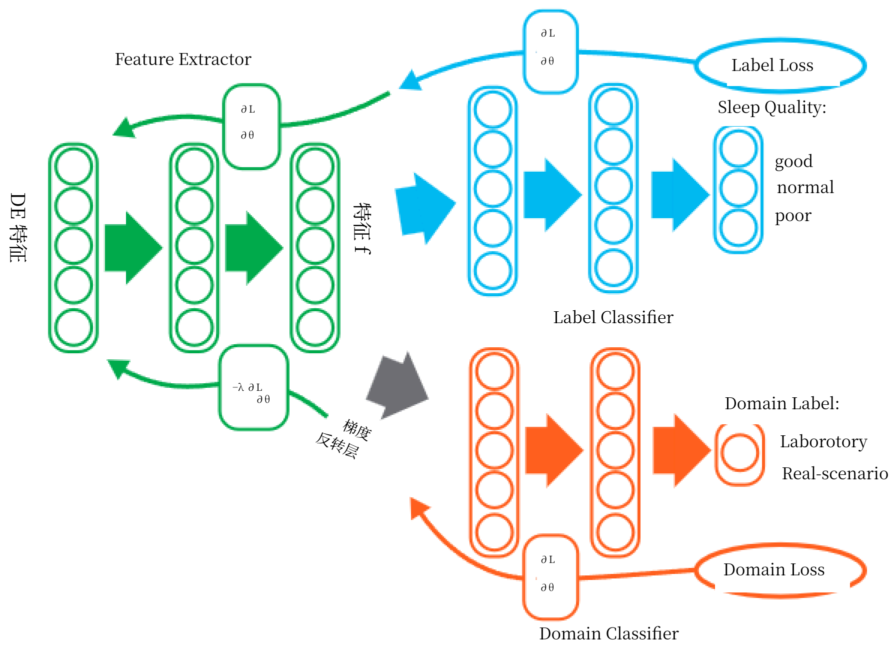
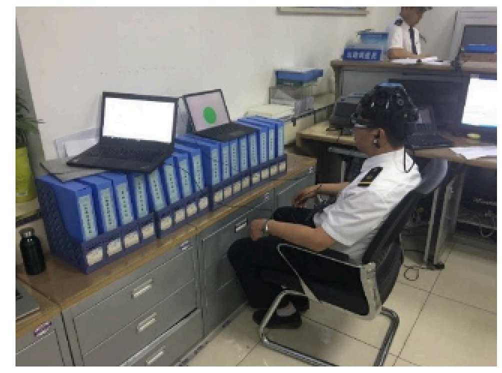


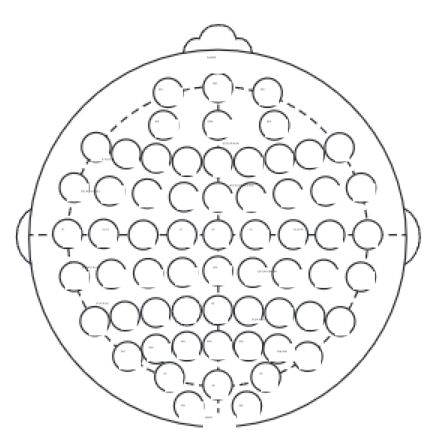
图 1. 具有三个组件的 DANN 模型：一个特征提取器 G，一个标签预测器 G 和一个域分类器 G。特征提取器将输入 x 映射到特征向量 f ∈ R，其所有层参数表示为θ，即 f = G(x; θ)。标签预测器将 f 映射到预测标签ˆ y，其所有层参数表示为θ。域分类器将 f 映射到预测域标记ˆd，参数表示为θ。

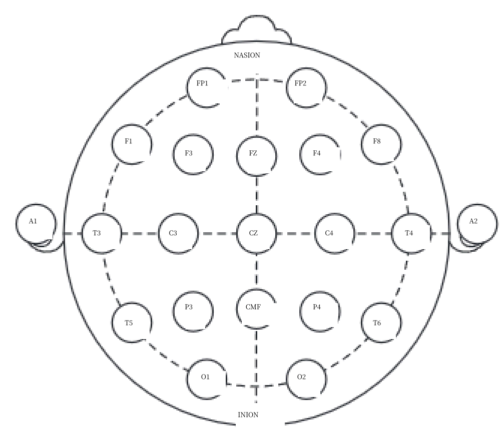
(a) 实验室环境，采集学生受试者的脑电图



(b) 真实场景实验环境，采集高速列车驾驶员的脑电图

(c) 湿电极脑电图帽 (d) 干电极 DSI-24 头戴式设备





(e) 更高分辨率的国际 10-20 系统 (f) 国际 10-20 系统 图 2. 实验细节（左侧列表示实验室实验，右侧列表示真实场景实验）。

IV. E 由于我们旨在将知识从实验室转移到实际场景，因此需要这两个场景的数据。本研究中使用的实验室数据是在我们之前关于睡眠质量估计的研究[7]中获得的，而在本研究中，我们从高速列车司机那里收集了实际场景数据。为了比较这两种类型的实验并易于理解实际场景数据与实验室数据的不同之处，我们将详细描述这两种数据。数据采集后进行预处理，并从预处理数据中提取 DE 特征。然后我们对提取的特征进行跨受试者和跨场景的任务。在这两个任务中，我们采用了基线 SVM 模型和八种 DA 方法。

A. 实验室睡眠实验

实验室实验旨在收集不同睡眠质量下的静息脑电图。每个实验包括两个部分：睡眠和脑电图采集。实验招募了 10 名受试者（六名男性，四名女性，年龄范围：21-26 岁，平均：23.57 岁，标准差：1.62 岁）。在脑电图采集前一晚，受试者被要求睡眠特定时间：4、6 或 8 小时，分别对应良好、正常或较差的睡眠质量[3]。为确保受试者按要求睡眠，他们佩戴了智能手环监测手腕运动，而人在入睡时很难检测到这种运动[7]。受试者醒来后，脑电图采集程序在不到一小时后开始，持续 30 分钟。脑电图信号使用 ESI NeuroScan 系统，通过 62 通道湿电极帽进行记录。帽上的电极按照

更高分辨率的国际 10-20 系统。图 2 的左侧列显示了实验场景和湿电极设备。在数据采集过程中，要求受试者注视屏幕上的一个绿点并默默计数，以保持平静和专注的状态[7]。

B. 真实场景睡眠实验

表 I

睡眠时长与相应睡眠质量

睡眠时间	睡眠质量	受试者
(2, 5] 小时	差	10
(5, 7] 小时	正常	24
(7, 10] 小时	优秀	36

真实场景实验用于记录特定组受试者的脑电图数据。受试者应进行日常活动，不受实验设定的任何条件限制。在本工作中，

we performed our experiment on the high-speed train drivers right before they start to work. 70 subjects (70 males, age range: 25-49, mean: 37.10, std: 4.69) are recruited. At the night before EEG acquisition, the sleep time of the subjects were not controlled. Before the experiments began, the subjects were required to fill out a questionnaire about their sleep last night. The questions investigated sleep duration, dreams and current feelings. In this work, we use the sleep duration as an index of subjects' sleep quality, as shown in Table I.

The EEG acquisition procedure is performed 30 minutes right before the subjects went to work. During EEG recording, the subject was required to stare at a green dot on the screen for 1 minute and close for another 1 minute alternately for 3 times. During the whole session, the subjects were required to count silently to keep a peaceful and concentrated state. The EEG signals were recorded with an 18-channel dry electrode cap using the DSI-24 device at a sampling rate of 300 Hz. Electrodes on the cap are placed according to the international 10-20 system. The experiment environment and dry electrode DSI-24 device are shown in the right column of Figure 2.

C. *Differences between experiments*

In order to keep the nature of the real scenario, instead of imposing restrictions on subjects, we adapt our experiment to the limitations in the real world, thus making the real-scenario experiments vary from the laboratory ones in many different ways. In order to keep our experiment short to fit the time limitation, we shrink our actual EEG recording time from 30 minutes to 6 minutes. In laboratory experiments, 64-channel wet electrode devices are used to record the EEG signals. However, these devices require laborious setup processes, therefore do not meet the demand of frequent acquisition of EEG signals in the real scenario. This demand is well addressed with the dry electrode device DSI-24 headset, which is more convenient to use. Moreover, we do not control the subjects' sleep time and experiment start time, so that the mental states of subjects during the experiments are close to those in their daily lives. For the ease of understanding, we summarized the difference between experiments in different scenarios as Table II.

D. *Preprocessing*

Some of the hardware induced difference can be eliminated by preprocessing methods. Preprocessed data in both scenarios would be sharing the same format and we can apply same subsequent processing procedures to them.

1) Electrodes Selection: In the laboratory, a wet electrodes cap with 62 channels was used and in the real scenario, a dry electrodes headset with 18 channels was used. The former one's electrodes are placed according to the higher-resolution international 10-20 system and the later according to the normal international 10-20 system. As a matter of fact, extra electrodes in higher resolution 10-20 system are added using the 10% division, which fills in intermediate sites halfway between those of the existing 10-20 system. In other words, if we remove the extra electrodes in the higher resolution system, the electrodes left would match perfectly with the normal system. We selected the corresponding electrodes from the laboratory device, as shown in Figure 3.

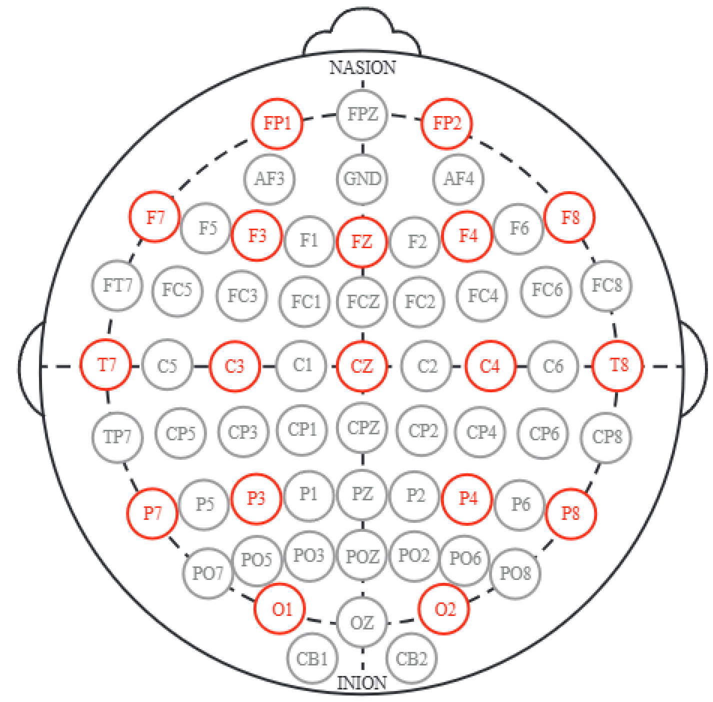


Fig. 3. Electrodes selection on laboratory device. Those electrodes marked red are selected ones mathching electrodes on DSI-24 headset. The CMF electrode on DSI-24 headset dose not record any signal, so the correponding PZ electrode is not marked.

2) Resetting Reference: As shown in Table II, the EEG data recorded with different devices have inconsistent references. The real-scenario data use the mean of A1 and A2 electrodes as references. However, the laboratory data use a specific REF electrode as the reference. Besides, the wet-electrode cap we used does not have any of the A1 or A2 electrodes, and also the DSI-24 headset does not have a REF electrode. To address the inconsistency, we decided to use the mean signal of all 18 electrodes as reference [7].

E. *Modeling*

We extract DE features for EEG signals in both scenarios with time window set to 1s. For the laboratory data, we collect EEG signals for 10 subjects with 3 experiments per subject and each experiment lasts about 30 minutes. Therefore, we have 58172 laboratory samples. For the real-scenario data, we collected EEG signals for 70 subjects with 1 experiment per subject and the valid time of each experiment is 3 minutes exactly. Therefore, we have 12600 real-scenario samples.

We mainly perform two modeling tasks in this work: (a) training and testing on real-world data, and (b) training on laboratory data but testing on real-world data. While performing the first task, the model performance is evaluated in a 5-fold cross validation manner. The real-world data is split into 5 parts, each containing 14 subjects and no subject's data should appear in multiple validation parts. Therefore, in the following discussion we refer to the first task as the **Cross-Subject** one and the second as the **Cross-Scenario** one. For the ease of comparison between two tasks, we split the real-scenario data into 5 parts in the same way as the cross-subject task while evaluating on the cross-scenario task.

After the feature extraction, We use the LIBLINEAR [24] SVM with default parameters as the baseline model on both

我们在高速列车司机开始工作前进行了实验。招募了 70 名受试者（70 名男性，年龄范围：25-49 岁，平均：37.10 岁，标准差：4.69 岁）。在脑电图采集前夜，受试者的睡眠时间未受控制。实验开始前，要求受试者填写一份关于昨晚睡眠情况的问卷。问卷调查了睡眠时长、梦境和当前感受。在本工作中，我们使用睡眠时长作为受试者睡眠质量的指标，如表 I 所示。

脑电图采集程序在受试者上班前 30 分钟进行。在脑电图记录期间，受试者被要求交替地盯着屏幕上的一个绿点 1 分钟，闭眼 1 分钟，共进行 3 次。在整个过程中，受试者被要求默默数数以保持平静和专注的状态。脑电图信号使用 DSI-24 设备，通过 18 通道的干电极帽以 300 Hz 的采样率进行记录。帽上的电极按照国际 10-20 系统放置。实验环境和干电极 DSI-24 设备显示在图 2 的右列。

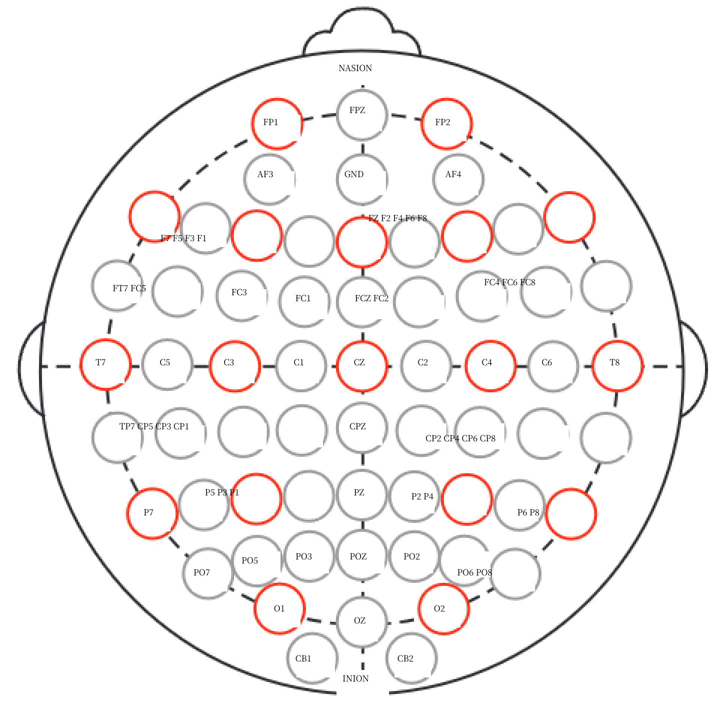


图 3. 实验室设备上的电极选择。红色标记的电极是选中的电极，与 DSI-24 头戴设备的电极相匹配。DSI-24 头戴设备上的 CMF 电极不记录任何信号，因此相应的 PZ 电极没有标记。

C. 实验差异

为了保持真实场景的特性，我们没有对受试者施加限制，而是根据现实世界的限制调整我们的实验，因此真实场景实验在许多方面与实验室实验有所不同。为了使实验时间适应时间限制，我们将实际的脑电图记录时间从 30 分钟缩短到 6 分钟。在实验室实验中，使用 64 通道湿电极设备记录脑电图信号。然而，这些设备需要繁琐的设置过程，因此无法满足在真实场景中频繁获取脑电图信号的需求。干电极设备 DSI-24 头戴式设备很好地解决了这一需求，它使用起来更加方便。此外，我们没有控制受试者的睡眠时间和实验开始时间，以便使实验期间受试者的精神状态更接近他们日常生活中的状态。为了便于理解，我们将不同场景下实验的差异总结为表 II。

D. 预处理

部分硬件引起的差异可以通过预处理方法消除。两种场景中的预处理数据将采用相同的格式，我们可以对它们应用相同的后续处理步骤。

1) 电极选择：在实验室中，使用了一个具有 62 通道的湿电极帽，而在真实场景中，使用了一个具有 18 通道的干电极头戴设备。前者电极的放置遵循高分辨率国际 10-20 系统，后者则遵循标准国际 10-20 系统。实际上，高分辨率 10-20 系统中的额外电极是通过 10%分法添加的，这些电极填补了现有 10-20 系统电极之间的中间位置。换句话说，如果我们移除高分辨率系统中的额外电极系统，留下的电极将完美匹配正常系统。我们从实验室设备中选择了相应的电极，如图 3 所示。

2) 重置参考：如表 II 所示，使用不同设备记录的脑电图数据具有不一致的参考。真实场景数据使用 A1 和 A2 电极的平均值作为参考。然而，实验室数据使用特定的 REF 电极作为参考。此外，我们使用的湿电极帽没有 A1 或 A2 电极，DSI-24 头戴设备也没有 REF 电极。为了解决这种不一致性，我们决定使用所有 18 个电极的平均信号作为参考[7]。

E. 建模

我们以 1 秒的时间窗口从两种场景中提取脑电图信号的 DE 特征。对于实验室数据，我们收集了 10 名受试者的脑电图信号，每位受试者进行 3 次实验，每次实验持续约 30 分钟。因此，我们有 58172 个实验室样本。对于真实场景数据，我们收集了 70 名受试者的脑电图信号，每位受试者进行 1 次实验，每次实验的有效时间为正好 3 分钟。因此，我们有 12600 个真实场景样本。

我们在这项工作中主要执行两项建模任务：(a) 在真实世界数据上进行训练和测试，以及(b) 在实验室数据上进行训练但在真实世界数据上进行测试。在执行第一项任务时，模型性能采用 5 折交叉验证的方式进行评估。真实世界数据被分成 5 份，每份包含 14 名受试者，且任何受试者的数据都不应出现在多个验证部分中。因此，在接下来的讨论中，我们将第一项任务称为跨受试者任务，第二项任务称为跨场景任务。为了便于比较两项任务，我们在评估跨场景任务时，以与跨受试者任务相同的方式将真实场景数据分成 5 份。

特征提取后，我们使用具有默认参数的 LIBLINEAR [24] SVM 作为基准模型，在两者上

TABLE II
DIFFERENCES BETWEEN LABORATORY AND REAL-SCENARIO EXPERIMENTS

Categories	Specifications	Laboratory	Real-scenario
Subjects	Age	21-26	25-49
	Sex-distribution	4 males, 6females	70 males
	Occupation	Students	High-speed train drivers
	Experiments per subject	3	1
Controlled conditions	Sleep time	4,6,8 hours	Not controlled
	Monitoring sleep	Yes	No
	Head Cleaning	Yes	No
	Experiment Starts	≤1 hour after wake up	1-10 hour after wake up
Experiment settings	Experiment duration	30 min	6 min
	Task	Eyes open	Eyes open and close
	Environment	Quite and isolated	Noisy
Devices	Electrods type	Wet electrodes	Dry electrodes
	Resolution	62 channels	18 channels
	Reference	REF(Between CPZ and PZ)	A1 and A2 (On the ears)
	Common Mode Follower	0	1 (CMF)
	Sample rate	1000 Hz	300 Hz

tasks. In the cross-subject task, while performing a 5-fold cross-validation, we view the fold held out as the target domain and the other 4 folds together as the source domain. In the cross-scenario task, we consider the whole real-world dataset as the target domain and the laboratory dataset as the source domain. TCA, ITL, GFK, JDA, SA, TJM and MIDA are run on all data as dimensionality reduction step and then SVM classifier is trained on transformed labeled source domain data and tested on the transformed target domain data. To boost the performance of these models, we pick the best subspace dimension parameter from the range $\{10,...,80\}$. The DANN model was trained with labeled source domain data and unlabeled target domain data, and the target domain labels can only be seen in evaluation steps. The tradeoff parameter λ of the GRL in DANN is set to 0.01. In the following discussions, we refer to the DA models except for DANN as the **baseline DA models** and refer to the baseline model as the **baseline SVM model**.

V. RESULTS AND DISCUSSION

We first compare DANN with baseline DA methods and also the baseline SVM model in terms of classification accuracy. The classification accuracies of cross-subject and cross-scenario models are summarized as Table III and IV.

DANN achieves much better performance than baseline DA models on both tasks. The average classification accuracy of DANN on cross-subject task is 73.72%. The performance improved 6.97% compared to MIDA (best baseline DA model), and 19.55% compared to the baseline SVM model. On the cross-scenario task, the accuracy of DANN model is 65.29%, which is 7.83% and 23.50% higher than the MIDA (best baseline DA model) and the baseline SVM model, respectively. We also observed that the standard deviation of DANN test accuracy across the 5 folds is lower than all the baseline DA models. Especially on the cross-scenario task, the lowest standard deviation of baseline DA models is 10.6%, while DANN achieves a standard deviation of 3.80%. This suggests that DANN is more stable than baseline DA models.

TABLE III
ACCURACY (%) OF CROSS-SUBJECT TASK

Method	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean±Std
SVM	59.57	48.41	57.42	55.71	49.74	54.17±4.87
TCA	58.37	72.98	63.77	69.48	53.85	63.69±7.82
ITL	66.07	66.79	57.14	76.43	56.71	64.63±8.13
GFK	59.41	66.79	53.73	58.45	58.49	59.37±4.70
JDA	59.41	59.52	**76.11**	60.44	55.91	62.28±7.92
SA	**77.42**	67.38	60.36	70.68	46.47	64.46±11.7
TJM	65.91	**74.72**	58.73	62.22	59.37	64.19±6.53
MIDA	71.35	69.76	66.91	69.21	56.51	66.75±5.94
DANN	67.50	74.64	71.23	**76.98**	**78.25**	**73.72**±4.38

TABLE IV
ACCURACY (%) OF CROSS-SCENARIO TASK

Method	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean±Std
SVM	43.86	40.35	42.18	38.98	43.59	41.79±2.10
TCA	54.60	61.19	38.89	67.34	56.87	55.60±10.6
ITL	53.69	63.10	24.68	60.71	49.41	50.32±13.7
GFK	49.64	62.46	40.99	**70.95**	49.25	54.66±11.9
JDA	47.70	61.43	35.71	66.11	66.94	55.58±13.5
SA	42.74	62.54	37.50	61.23	54.17	51.64±11.1
TJM	41.67	60.28	38.97	59.88	60.60	52.28±10.9
MIDA	56.47	**66.35**	38.69	68.93	56.87	57.46±11.9
DANN	**69.10**	64.68	**60.79**	62.6296	**69.25**	**65.29**±3.80

We can also see that all the transfer learning methods achieved a better classification accuracy than the baseline SVM model on both tasks. With domain adaptation, the mean accuracy is improved at least 5.20% and 9.85% on cross-subject and cross-scenario, respectively. We can see that the DA methods do eliminate the difference between the source and target domain. However, the major limitation of baseline DA models is that they use predefined functionalities to measure the domain difference and transform the data in a

表 II

实验室与现实场景实验的差异

类别 规格 实验室 现实场景	
受试者	年龄 21-26 25-49 性别分布 4 名男性，6 名女性 70 名男性 职业 学生 高速列车司机 每个受试者的实验次数 3 1
控制条件	睡眠时间 4,6,8 小时 不控制 监测睡眠 是 否 头部清洁 是 否 实验开始 睡醒后 ≤1 小时 睡醒后 1-10 小时
实验设置	实验时长 30 分钟 6 分钟 任务 睁眼 睁眼和闭眼 环境 安静且隔离 嘈杂
设备	电极类型 湿电极 干电极 分辨率 62 通道 18 通道 参考电极 REF（CPZ 和 PZ 之间） A1 和 A2（在耳朵上） 共模跟随器 0 1（CMF） 采样率 1000Hz 300Hz

任务。在跨主题任务中，进行 5 折交叉验证时，我们将保留的折视为目标域，其余 4 折视为源域。在跨场景任务中，我们将整个真实世界数据集视为目标域，实验室数据集视为源域。TCA、ITL、GFK、JDA、SA、TJM 和 MIDA 对所有数据进行降维处理，然后使用转换后的源域标记数据进行 SVM 分类器训练，并在转换后的目标域数据上进行测试。为了提升这些模型的性能，我们从范围{10,...,80}中选择最佳子空间维度参数。DANN 模型使用标记的源域数据和未标记的目标域数据进行训练，目标域标签仅在评估步骤中可见。DANN 中 GRL 的权衡参数λ设置为 0.01。在后续讨论中，我们将除 DANN 以外的 DA 模型称为基线 DA 模型，将基线模型称为基线 SVM 模型。

V. 结果与讨论

我们首先在分类准确率方面比较了 DANN 与基线 DA 方法以及基线 SVM 模型。跨被试和跨场景模型的分类准确率总结如表 III 和 IV 所示。

DANN 在两项任务上的性能都远优于基线 DA 模型。DANN 在跨被试任务上的平均分类准确率为 73.72%。与 MIDA（最佳基线 DA 模型）相比，性能提高了 6.97%；与基线 SVM 模型相比，性能提高了 19.55%。在跨场景任务上，DANN 模型的准确率为 65.29%，分别比 MIDA（最佳基线 DA 模型）和基线 SVM 模型高 7.83%和 23.50%。我们还观察到，DANN 在 5 折测试中的准确率标准差低于所有基线 DA 模型。特别是在跨场景任务上，基线 DA 模型的标准差最低为 10.6%，而 DANN 的标准差为 3.80%。这表明 DANN 比基线 DA 模型更稳定。

表 III

A(%) C-ST

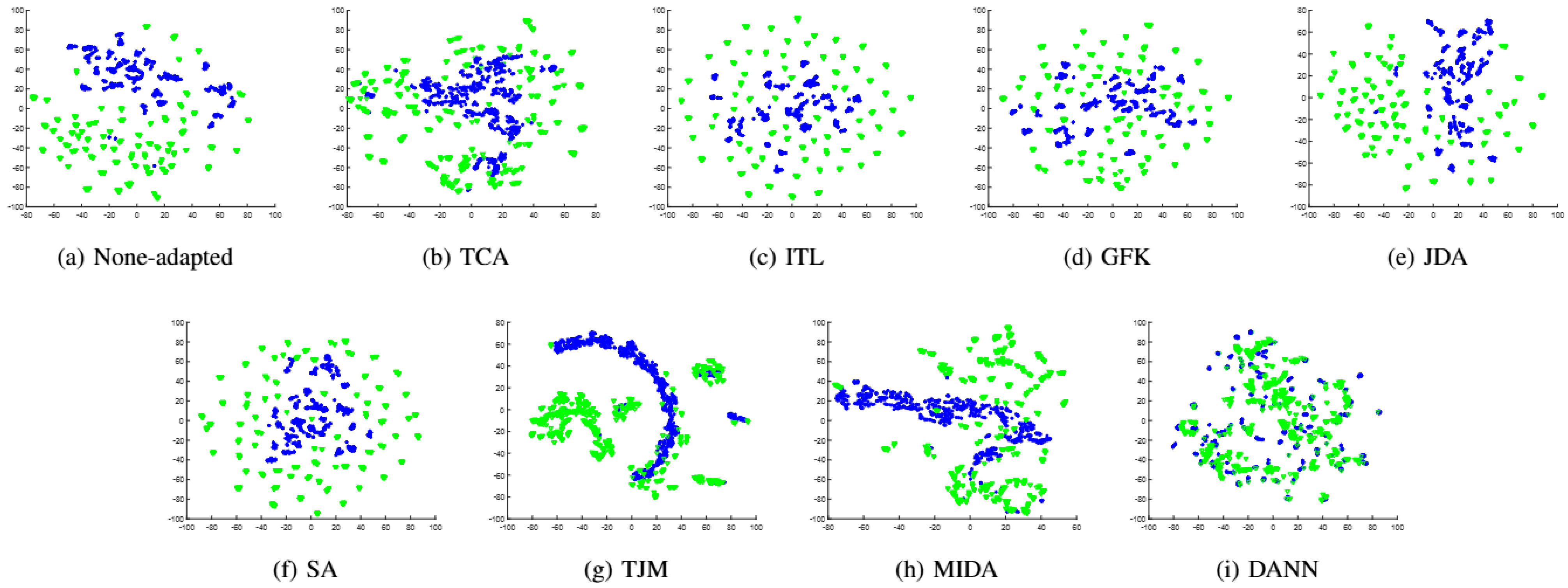
方法 折叠 1 折叠 2 折叠 3 折叠 4 折叠 5 平均±标准差
SVM 59.57 48.41 57.42 55.71 49.74 54.17±4.87 TCA 58.37 72.98 63.77 69.48 53.85 63.69±7.82 ITL 66.07 66.79 57.14 76.43 56.71 64.63±8.13 GFK 59.41 66.79 53.73 58.45 58.49 59.37±4.70 JDA 59.41 59.52 76.11 60.44 55.91 62.28±7.92 SA 77.42 67.38 60.36 70.68 46.47 64.46±11.7 TJM 65.91 74.72 58.73 62.22 59.37 64.19±6.53 MIDA 71.35 69.76 66.91 69.21 56.51 66.75±5.94 DANN 67.50 74.64 71.23 76.98 78.25 73.72±4.38

表 IV

A(%) C-ST

方法 折叠 1 折叠 2 折叠 3 折叠 4 折叠 5 Mean±Std
SVM 43.86 40.35 42.18 38.98 43.59 41.79±2.10 TCA 54.60 61.19 38.89 67.34 56.87 55.60±10.6 ITL 53.69 63.10 24.68 60.71 49.41 50.32±13.7 GFK 49.64 62.46 40.99 70.95 49.25 54.66±11.9 JDA 47.70 61.43 35.71 66.11 66.94 55.58±13.5 SA 42.74 62.54 37.50 61.23 54.17 51.64±11.1 TJM 41.67 60.28 38.97 59.88 60.60 52.28±10.9 MIDA 56.47 66.35 38.69 68.93 56.87 57.46±11.9 DANN 69.10 64.68 60.79 62.6296 69.25 65.29±3.80

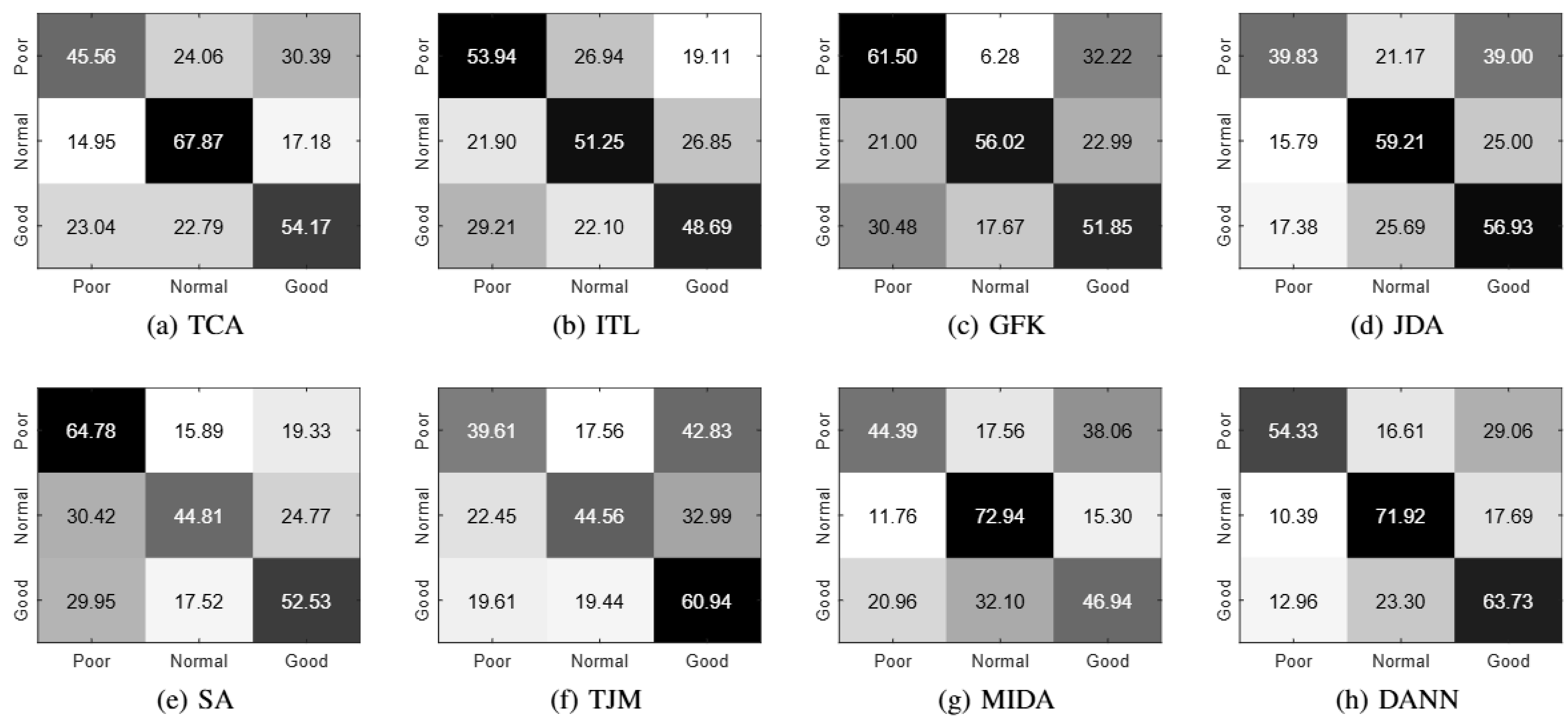
我们也可以看到，所有迁移学习方法在两个任务上都取得了比基线 SVM 模型更好的分类准确率。通过领域自适应，交叉受试者（crosssubject）和跨场景（cross-scenario）的平均准确率分别提高了至少 5.20%和 9.85%。我们可以看到，领域自适应（DA）方法确实消除了源域和目标域之间的差异。然而，基线 DA 模型的主要局限性在于它们使用预定义的功能来测量领域差异，并转换数据。



(a) None-adapted (b) TCA (c) ITL (d) GFK (e) JDA

(f) SA (g) TJM (h) MIDA (i) DANN

Fig. 4. The effects of adaptation on cross-scenario task (Best viewed in color). The figure shows t-SNE [25] visualizations of features. (a) shows features with no adaptation; (b)-(h) show features after transformation by baseline DA methods; (i) shows the feature transformed by the feature extractor of DANN. Blue dots correspond to the source domain examples, while green triangles represent the target domain ones. The source and target domain features after transformation are much closer in DANN than baseline DA models



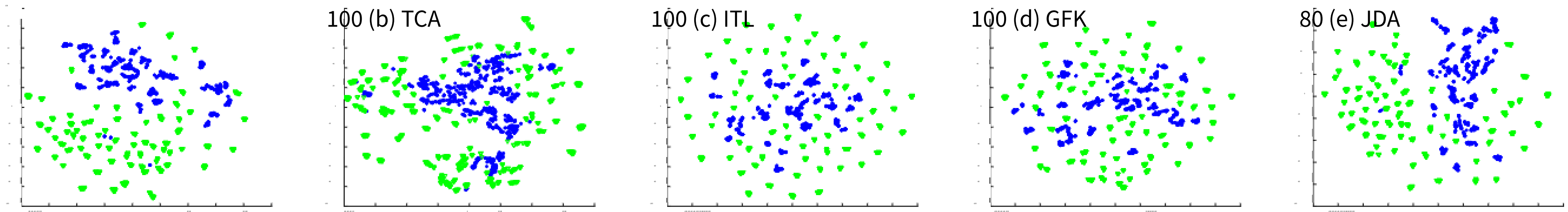
(a) TCA (b) ITL (c) GFK (d) JDA

(e) SA (f) TJM (g) MIDA (h) DANN

Fig. 5. The confusion matrix of DA models on cross-scenario task. The vertical axis represents the labels marked by sleep duration according to Table I; The horizontal axis represents the predicted label of each DA model. The number on each block reprensents the percentage of samples with that the true label (vertical ones) being classified to the predicted label (horizontal ones).

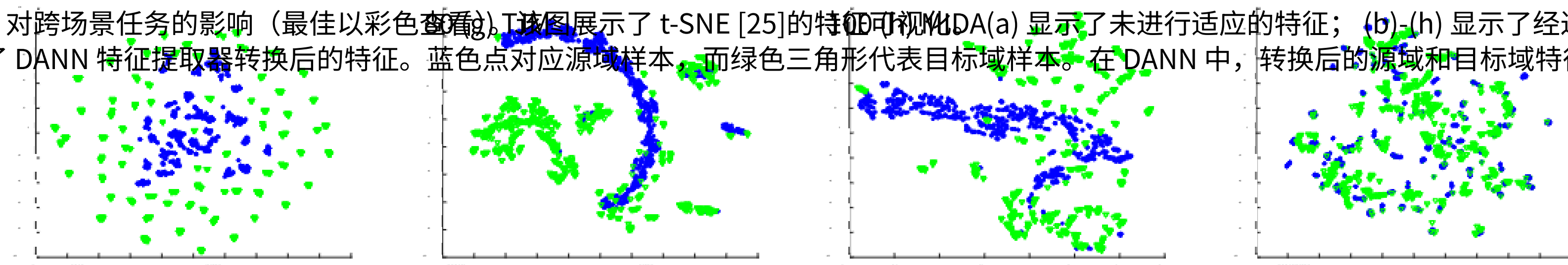
predetermined way, which is rather rigid compared to the DANN method. DANN represents both the domain difference and the transformation with multilayer neural networks and can be trained in an adversarial manner with SGD based methods, which makes it easier to reduce the domain discrepancy between source and target domains. Figure 4 shows how DANN succeeded at aligning feature distributions on the cross-scenario task. Another drawback of the insufficient adaptation of baseline DA methods is that it may cause negative transfer in some components of the target domain. We can observe this phenomenon in Table IV, where negative transfer occurs in Fold 3 on TCA, ITL, JDA, SA and TJM models, and yet DANN shows no negative transfer in all 5 Folds.

It is noteworthy that the real-scenario data is severely unbalanced, with 1800 poor-sleep-quality samples, 4320 normal-sleep-quality samples and 6480 good-sleep-quality samples. Thus the classifiers may benefit from biased predictions. For a simple example, if one classifies all samples to be of good sleep quality, it would achieve over 50% classification accuracy. With that in mind, we visualize the confusion matrix of each model on cross-scenario task to see whether the models have made biased predictions (see Figure 5). We observed that TJM and JDA do suffer from the biased prediction problem. TJM achieved a classification accuracy of 60.94% on good-sleep-quality samples while its accuracy drops sharply to 39.61% on poor-sleep quality-samples, which is close to

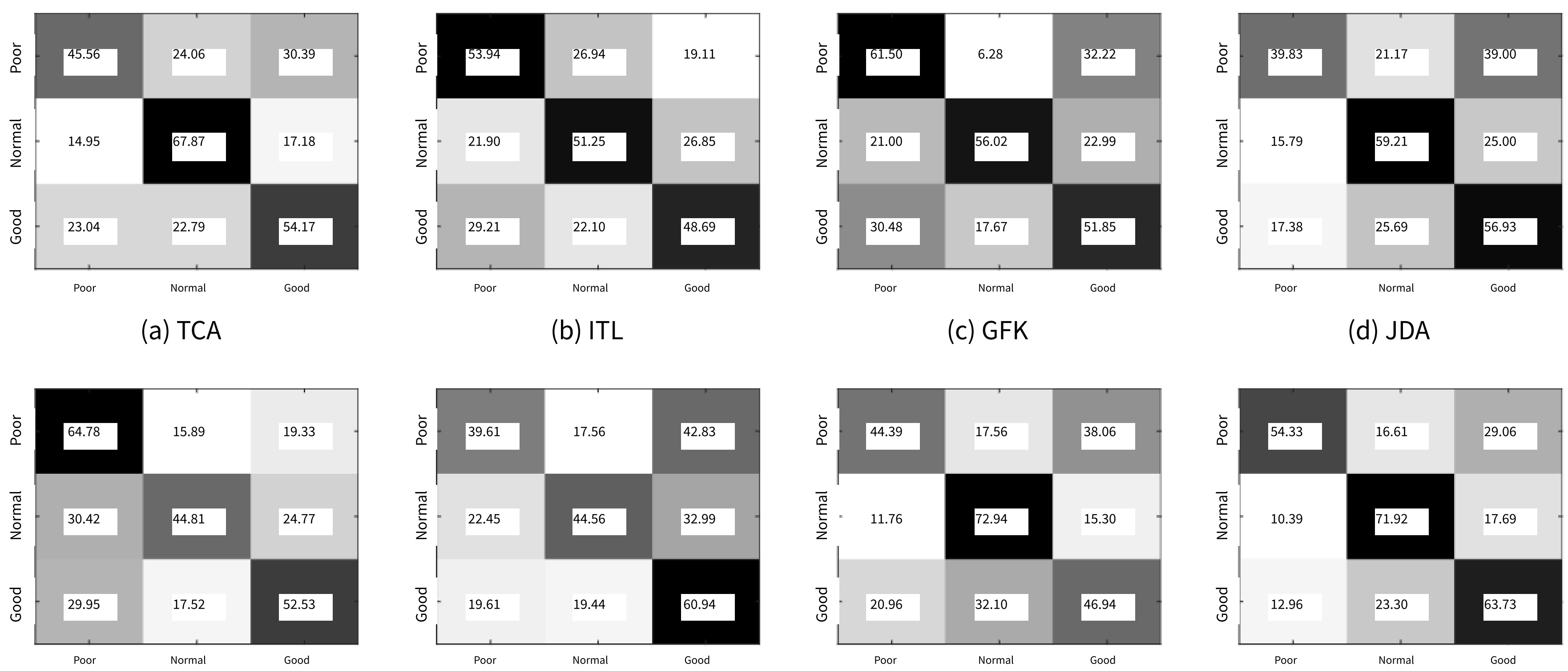


(a) 未适配

100 (i) DANN 图 4. 对跨场景任务的影响（最佳以彩色查看）。(g) TJM (h) MIDA 该图展示了 t-SNE [25]的特征可视化。(a) 显示了未进行适应的特征；(b)-(h) 显示了经过基线 DA 方法转换后的特征；(i) 显示了 DANN 特征提取器转换后的特征。蓝色点对应源域样本，而绿色三角形代表目标域样本。在 DANN 中，转换后的源域和目标域特征比基线 DA 模型更接近。



(f) SA



(a) TCA (b) ITL (c) GFK (d) JDA

(e) SA (f) TJM (g) MIDA (h) DANN

图 5. DA 模型在跨场景任务上的混淆矩阵。垂直轴表示根据表 I 标记的睡眠时长标签；水平轴表示每个 DA 模型的预测标签。每个块上的数字表示具有该真实标签（垂直轴）被分类到预测标签（水平轴）的样本百分比。

预定的方法，与 DANN 方法相比较为僵化。DANN 既能表示域差异，又能通过多层神经网络进行转换，并可以基于 SGD 方法进行对抗性训练，这使得它更容易减少源域和目标域之间的域差异。图 4 显示了 DANN 如何成功在跨场景任务上对齐特征分布。基线 DA 方法适应不足的另一个缺点是可能导致目标域某些组件出现负迁移。我们可以在表 IV 中观察到这种现象，其中 TCA、ITL、JDA、SA 和 TJM 模型在 Fold 3 上出现负迁移，而 DANN 在所有 5 个 Fold 中均未出现负迁移。

值得注意的是，真实场景数据严重不平衡，其中包含 1800 个睡眠质量差样本、4320 个睡眠质量正常样本和 6480 个睡眠质量好样本。因此，分类器可能会受益于有偏见的预测。以一个简单的例子来说，如果将所有样本都分类为睡眠质量好，那么分类准确率将超过 50%。基于这一点，我们在跨场景任务中可视化了每个模型的混淆矩阵，以查看模型是否进行了有偏见的预测（见图 5）。我们观察到 TJM 和 JDA 确实存在有偏见的预测问题。TJM 在睡眠质量好样本上的分类准确率达到 60.94%，而其在睡眠质量差样本上的准确率急剧下降至 39.61%，接近

a random prediction result. In contrast, DANN's prediction preserves a 54.33% classification accuracy on poor-sleep-quality samples.

Lastly, we compare the performance of models on different tasks. All models perform with a relatively lower classification accuracy on the cross-scenario task compared to the cross-subject task, which is predictable as the former is more challenging. As we have illustrated in Table II, the cross-scenario task is dealing with the reality gap constituted with differences of four categories. The intersubject variation that the cross-subject task focuses on is merely one of these categories. Therefore, the baseline SVM model performs with a low accuracy (41.79%) on cross-scenario task. In spite of the difficulties, DANN achieves remarkable improvement over baseline DA methods, which shows that DANN performs more robustly on difficult domain adaptation tasks than baseline DA models.

VI. Conclusion

We have studied last-night sleep quality estimation in real scenarios. To acquire the real-scenario data, we have performed EEG acquisition for high-speed train drivers without changing their work schedules. In order to meet the demand of practical application, we have introduced a dry electrode device to collect the EEG signals. To estimate the sleep quality of the drivers, we have evaluated two tasks: the cross-subject one and the cross-scenario one. Eight DA methods have been adopted to both tasks to reduce the domain discrepancy between source and target domains. In this work, we have adopted a novel DANN approach, which is based on neural networks with an adversarial architecture. The results have shown that DANN approach considerably outperforms baseline models on both tasks in terms of classification accuracy. Moreover, we have also analyzed the models from various perspectives and concluded that the DANN approach is also stable and robust on the challenging cross-scenario task.

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References

[1] D. J. Buysse, C. F. Reynolds, T. H. Monk, S. R. Berman, and D. J. Kupfer, "The pittsburgh sleep quality index: A new instrument for psychiatric practice and research," *Psychiatry Research*, vol. 28, no. 2, pp. 193 – 213, 1989.

[2] T. Akerstedt, "Consensus statement: fatigue and accidents in transport operations," *Journal of Sleep Research*, vol. 9, no. 4, pp. 395–395, 2000.

[3] L.-L. Wang, W.-L. Zheng, H.-W. Ma, and B.-L. Lu, "Measuring sleep quality from EEG with machine learning approaches," in *the IEEE International Joint Conference on Neural Networks*, 2016, pp. 905–912.

[4] D. J. Buysse, C. F. Reynolds, T. H. Monk, S. R. Berman, and D. J. Kupfer, "The pittsburgh sleep quality index: a new instrument for psychiatric practice and research," *Psychiatry Research*, vol. 28, no. 2, pp. 193–213, 1989.

[5] M. W. Johns, "A new method for measuring daytime sleepiness: the epworth sleepiness scale," *Sleep*, vol. 14, no. 6, pp. 540–545, 1991.

[6] Y.-K. Teng, L.-C. Chiang, K.-H. Lue, S.-W. Chang, L. Wang, S.-P. Lee, H. Ting, and S.-D. Lee, "Poor sleep quality measured by polysomnography in non-obese asthmatic children with or without moderate to severe obstructive sleep apnea," *Sleep Medicine*, vol. 15, no. 9, pp. 1062–1067, 2014.

[7] X.-Z. Zhang, W.-L. Zheng, and B.-L. Lu, "EEG-based sleep quality evaluation with deep transfer learning," in *International Conference on Neural Information Processing*. Springer, 2017, pp. 543–552.

[8] R. D. Ogilvie, "The process of falling asleep," *Sleep Medicine Reviews*, vol. 5, no. 3, pp. 247–270, 2001.

[9] E. A. Wolpert, "A manual of standardized terminology, techniques and scoring system for sleep stages of human subjects." *Archives of General Psychiatry*, vol. 20, no. 2, pp. 246–247, 1969.

[10] N. Li, Y. Wang, M. Wang, and H. Liu, "Effects of sleep deprivation on gamma oscillation of waking human EEG," *Progress in Natural Science*, vol. 18, no. 12, pp. 1533–1537, 2008.

[11] S. H. Na, S.-H. Jin, and S. Y. Kim, "The effects of total sleep deprivation on brain functional organization: mutual information analysis of waking human EEG," *International Journal of Psychophysiology*, vol. 62, no. 2, pp. 238–242, 2006.

[12] P. Tassi, A. Bonnefond, O. Engasser, A. Hoeft, R. Eschenlauer, and A. Muzet, "EEG spectral power and cognitive performance during sleep inertia: the effect of normal sleep duration and partial sleep deprivation," *Physiology & Behavior*, vol. 87, no. 1, pp. 177–184, 2006.

[13] K. Mann, P. Bäcker, and J. Röschke, "Dynamical properties of the sleep EEG in different frequency bands," *International Journal of Neuroscience*, vol. 73, no. 3-4, pp. 161–169, 1993.

[14] L.-C. Shi, Y.-Y. Jiao, and B.-L. Lu, "Differential entropy feature for EEG-based vigilance estimation," in *35th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2013, pp. 6627–6630.

[15] X.-W. Wang, D. Nie, and B.-L. Lu, "Emotional state classification from EEG data using machine learning approach," *Neurocomputing*, vol. 129, pp. 94–106, 2014.

[16] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Transactions on Neural Networks*, vol. 22, no. 2, p. 199, 2011.

[17] G. Niu, B. Dai, M. Yamada, and M. Sugiyama, "Information-theoretic semi-supervised metric learning via entropy regularization," in *29th International Conference on Machine Learning*, July 2012, pp. 89–96.

[18] B. Gong, Y. Shi, F. Sha, and K. Grauman, "Geodesic flow kernel for unsupervised domain adaptation," in *the IEEE Conference on Computer Vision and Pattern Recognition*, June 2012, pp. 2066–2073.

[19] M. Long, J. Wang, G. Ding, J. Sun, and P. S. Yu, "Transfer feature learning with joint distribution adaptation," in *the IEEE International Conference on Computer Vision*, 2013, pp. 2200–2207.

[20] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *the IEEE International Conference on Computer Vision*, 2013, pp. 2960–2967.

[21] M. Long, J. Wang, G. Ding, J. Sun, and P. S. Yu, "Transfer joint matching for unsupervised domain adaptation," in *the IEEE Conference on Computer Vision and Pattern Recognition*, 2014, pp. 1410–1417.

[22] K. Yan, L. Kou, and D. Zhang, "Learning domain-invariant subspace using domain features and independence maximization," *IEEE Transactions on Cybernetics*, vol. 48, no. 1, pp. 288–299, Jan 2018.

[23] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," in *the International Conference on Machine Learning*, 2015, pp. 1180–1189.

[24] R.-E. Fan, K.-W. Chang, C.-J. Hsieh, X.-R. Wang, and C.-J. Lin, "LIBLINEAR: A library for large linear classification," *Journal of Machine Learning Research*, vol. 9, no. Aug, pp. 1871–1874, 2008.

[25] D. Vazquez, A. M. Lopez, J. Marin, D. Ponsa, and D. Geronimo, "Virtual and real world adaptation for pedestrian detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 36, no. 4, pp. 797–809, 2014.

一个随机的预测结果。相比之下，DANN 的预测在睡眠质量差的样本上保持了 54.33%的分类准确率。

最后，我们比较了模型在不同任务上的性能。所有模型在跨场景任务上的分类准确率相对较低，与跨被试任务相比，这是可以预料的，因为前者更具挑战性。正如我们在表 II 中所展示的，跨场景任务处理的是由四个类别差异构成的现实差距。跨被试任务所关注的被试间差异仅是其中之一。因此，基线 SVM 模型在跨场景任务上的准确率较低（41.79%）。尽管存在困难，DANN 在基线 DA 方法上取得了显著改进，这表明 DANN 在困难的领域适应任务上比基线 DA 模型表现更鲁棒。

VI. C 我们研究了在真实场景下对昨晚睡眠质量的评估。为了获取真实场景数据，我们未改变高速列车驾驶员的工作安排，对其实施了脑电图采集。为了满足实际应用需求，我们引入了干电极设备来收集脑电图信号。为了评估驾驶员的睡眠质量，我们评估了两个任务：跨被试任务和跨场景任务。我们采用了八种域适应方法来这两个任务，以减少源域和目标域之间的域差异。在这项工作中，我们采用了一种基于具有对抗架构的神经网络的创新深度对抗神经网络方法。结果表明，在分类准确率方面，深度对抗神经网络方法在两个任务上都显著优于基线模型。此外，我们还从不同角度分析了模型，并得出结论：深度对抗神经网络方法在具有挑战性的跨场景任务上也是稳定和鲁棒的。

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[1] D. J. Buysse, C. F. Reynolds, T. H. Monk, S. R. Berman, and D. J. Kupfer，“匹兹堡睡眠质量指数：一种新的精神病学实践和研究工具”，《精神病学研究》，第 28 卷，第 2 期，第 193-213 页，1989 年。

[2] T. Akerstedt, “共识声明：运输操作中的疲劳和事故，” 《睡眠研究杂志》, 第 9 卷, 第 4 期, 第 395 页, 2000. [3] L.-L. Wang, W.-L. Zheng, H.-W. Ma, 和 B.-L. Lu, “基于机器学习方法从脑电图测量睡眠质量，” 在 IEEE 国际联合神经网络会议, 2016, 第 905-912 页.

[4] D. J. Buysse, C. F. Reynolds, T. H. Monk, S. R. Berman, and D. J. Kupfer，“匹兹堡睡眠质量指数：一种新的精神病学实践和研究工具”，《精神病学研究》，第 28 卷，第 2 期，第 193-213 页，1989 年。

[5] M. W. Johns, “一种新的白天嗜睡测量方法：爱泼沃斯嗜睡量表,”《睡眠》杂志, 第 14 卷, 第 6 期, 第 540-545 页, 1991. [6] Y.-K. Teng, L.-C. Chiang, K.-H. Lue, S.-W. Chang, L. Wang, S.-P. Lee, H. Ting, 和 S.-D. Lee, “通过多导睡眠图测量非肥胖哮喘儿童睡眠质量的研究——伴有或不伴有中重度阻塞性睡眠呼吸暂停,”《睡眠医学》杂志, 第 15 卷, 第 9 期, 第 1062-1067 页, 2014. [7] X.-Z. Zhang, W.-L. Zheng, 和 B.-L. Lu, “基于脑电图深度迁移学习的睡眠质量评估，” 在国际神经信息处理会议论文集中. 斯普林格出版社, 2017, 第 543-552 页. [8] R. D. Ogilvie, “入睡的过程，” 《睡眠医学综述》杂志, 第 5 卷, 第 3 期, 第 247-270 页, 2001. [9] E. A. Wolpert, “人类睡眠阶段标准化术语、技术和评分系统的手册.” 《美国精神病学杂志》档案, 第 20 卷, 第 2 期, 第 246-247 页, 1969.

[10] N. Li, Y. Wang, M. Wang, 和 H. Liu, “睡眠剥夺的影响” 人类清醒时脑电波的γ振荡，”自然科学进展，第 18 卷，第 12 期，第 1533-1537 页，2008 年。

[11] S.H. Na, S.-H. Jin, 和 S.Y. Kim, “总睡眠剥夺的影响” 关于大脑功能组织：清醒人类脑电图互信息分析，”国际心理生理学杂志，第 62 卷，第 2 期，第 238-242 页，2006 年。

[12] P. Tassi, A. Bonnefond, O. Engasser, A. Hoeft, R. Eschenlauer, 和 A. Muzet, “睡眠惯性期间脑电图频谱功率与认知表现：正常睡眠时长和部分睡眠剥夺的影响，” 生理学与行为学, 卷. 87, 第. 1, 页. 177–184, 2006.

[13] K. Mann, P. B¨acker, and J. R¨oschke, “Dynamical properties of the 不同频段睡眠脑电图，”国际神经科学杂志，第 73 卷，第 3-4 期，第 161-169 页，1993 年。

[14] L.-C. Shi, Y.-Y. Jiao, and B.-L. Lu, “差异熵特征 基于脑电图（EEG）的警觉性估计，”在 2013 年 IEEE 工程医学与生物学学会第 35 届年会上，第 6627-6630 页。

[15] 王晓伟, 聂丹, 和 陆宝林, “情绪状态分类从 使用机器学习方法处理脑电图数据，”神经计算，第 129 卷，第 94-106 页，2014 年。

[16] S. J. Pan, I. W. Tsang, J. T. Kwok 和 Q. Yang，“通过对抗域适应” 迁移组件分析，”IEEE 神经网络汇刊，第 22 卷，第 2 期，第 199 页，2011 年。

[17] G. Niu, B. Dai, M. Yamada, 和 M. Sugiyama, “基于信息的” 基于熵正则化的半监督度量学习，”在 2012 年 29 届国际机器学习会议，2012 年 7 月，第 89-96 页。

[18] B. Gong, Y. Shi, F. Sha, and K. Grauman, “Geodesic flow kernel for “无监督域适应，”在 2012 年 IEEE 计算机视觉与模式识别会议，第 2066-2073 页。

[19] M. Long, J. Wang, G. Ding, J. Sun, 和 P. S. Yu, “迁移特征 “基于联合分布适应的学习”，在 2013 年 IEEE 国际计算机视觉会议上，第 2200-2207 页。

[20] B. Fernando, A. Habrard, M. Sebban 和 T. Tuytelaars, “无监督 使用子空间对齐的视觉领域自适应，”在 2013 年 IEEE 国际计算机视觉会议，第 2960-2967 页。

[21] M. Long, J. Wang, G. Ding, J. Sun, 和 P. S. Yu, “迁移联合 在“无监督域适应的匹配”，2014 年 IEEE 计算机视觉与模式识别会议，第 1410-1417 页。

[22] K. Yan, L. Kou, and D. Zhang, “学习领域不变子空间 利用领域特征和独立性最大化，”IEEE 控制论汇刊，第 48 卷，第 1 期，第 288-299 页，2018 年 1 月。

[23] Y. Ganin 和 V. Lempitsky，“无监督域适应” 反向传播，”在 2015 年机器学习国际会议，第 1180-1189 页。

[24] R.-E. Fan, K.-W. Chang, C.-J. Hsieh, X.-R. Wang, 和 C.-J. Lin, “LIBLINEAR: 一个用于大规模线性分类的库，”机器学习研究杂志，第 9 卷，第 8 期，第 1871–1874 页，2008 年。

[25] D. Vazquez, A. M. Lopez, J. Marin, D. Ponsa, 和 D. Geronimo, “虚拟 和现实世界的行人检测适应，”模式分析和机器智能 IEEE 汇刊，第 36 卷，第 4 期，第 797– 809 页，2014 年。